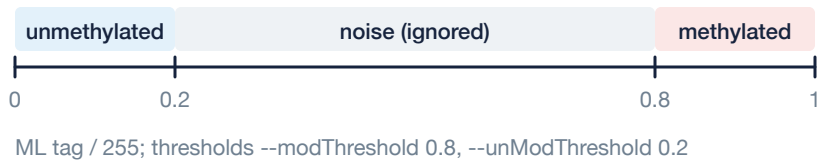


a Per-read CpG state from ML probability



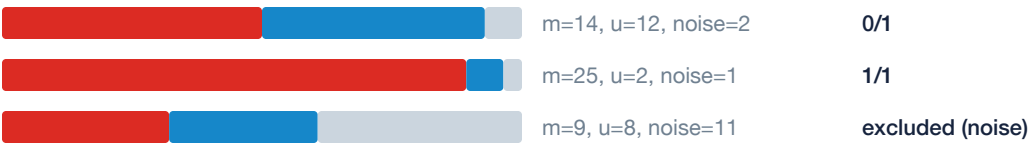
b CpG strand pairing and site genotype

5' ... C G ... 3' forward-strand reads report C; reverse-strand reads report the G position
→ reverse-strand calls are mapped to the forward CpG coordinate and the two strands are pooled

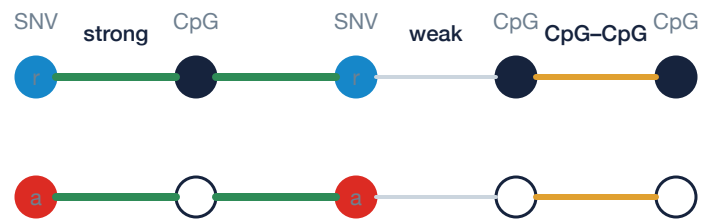
m = methylated reads, u = unmethylated reads, noise = depth – m – u

heterozygous (allele-specific) if $\min(m,u)/\max(m,u) \geq 0.6$ and $\text{noise}/\text{depth} \leq 0.2$

otherwise homozygous methylated ($m \geq u$) or unmethylated → excluded from phasing



c Refinement by co-segregation with SNVs



strong anchor: CpG state co-segregates with an SNV within the next 20 variants on $> \max(6, (d_1+d_2)/4)$ reads with majority ratio ≥ 0.9

weak CpG: no qualifying SNV link; accepted only if it co-segregates with an already accepted CpG (iterative expansion, 2 rounds)

adjacent CpG coordinates are merged and represented by the first

d Output VCF record

```
chr1 10469 . N . . PASS RS=P;MR=read3,read7,...;NR=read1,read4,...  
GT:MD:UD:DP 0/1:14:12:28
```

RS strand; MR/NR methylated and unmethylated read names (used by phase to attach 5mC alleles to reads); MD/UD/DP depths. Phased output adds PS and 0|1.